

C.J. Ouseph, Ph.D.

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PROFESSIONAL SUMMARY

Theoretical physicist with a Ph.D. in Physics and 5+ years of research and tertiary teaching experience in particle physics, cosmology and astrophysics. Author of 15+ peer-reviewed publications in internationally refereed journals, with research spanning primordial black holes, inflation, gravitational waves and new-physics phenomenology. Experienced in teaching Electrodynamics and Mathematical Physics, supervising Master's research students through publication, and developing computational tools to support physics education. Research programme is computationally driven and well suited to interdisciplinary collaboration across theoretical physics, astrophysics and space science.

RESEARCH SKILLS ALIGNED TO LECTURER IN PHYSICS REQUIREMENTS

- **A developing record of scholarly research and publication:** 15+ peer-reviewed articles in internationally refereed Q1 journals (JHEP, Physical Review D, European Physical Journal C, Journal of Physics G) arising from 16 HPC-based modelling projects in collider phenomenology and early-universe cosmology — an established publication record at the level expected of an early-career Lecturer.
- **A research agenda extendable to an Aotearoa New Zealand base:** Current programme covers primordial black holes, gravitational-wave signatures, inflationary dynamics, dark photons, axion-like particles and sterile neutrinos — theory-led and computational work requiring only Python, Mathematica and HPC access, so it transfers to a new institution without laboratory infrastructure and complements the School's existing astrophysics and theoretical physics strengths.
- **Demonstrated evidence of successful teaching at tertiary level:** Delivered lectures and tutorials for Electrodynamics (2020–2021) and Mathematical Physics (2022–2023) at National Tsing Hua University — two 2-hour classes per week, full syllabus coverage from Griffiths and Jackson, prepared lecture notes and problem sets, marked and moderated assignments and examination scripts; supplementary materials written to bridge students' mathematical prerequisites produced a measured rise in assignment scores.
- **Supervising student research at Honours, Masters and/or PhD level:** Supervised two Master's students and one exchange student at National Tsing Hua University through problem formulation, statistical analysis and scientific writing, each project completing to co-authored publication in JHEP (2024) and Physical Review D (2023).
- **Engagement and collaboration with colleagues in teaching and research:** Led research collaborations spanning five countries and organised the NTHU Physics Department seminar series and Master's/PhD workshop series in Fall 2023, inviting external speakers and arranging facilities — the same collegial coordination the School's teaching and research programmes rely on.
- **Commitment to technology and innovation in teaching:** Built Inflation Explorer, an interactive Plotly Dash dashboard that lets students sweep inflationary potential parameters and watch n_s and r move against Planck 2018 and BICEP/Keck constraint bands, and VizDash Pro, a browser-based statistical visualisation studio — both designed as teaching aids as much as research tools.

TEACHING & MENTORSHIP EXPERIENCE

- **National Tsing Hua University** Hsinchu, Taiwan
Teaching Assistant — Electrodynamics (P.h.D., Postgraduate & Undergraduate Physics) 2020 - 2021
 - **Lecture & Tutorial Delivery:** Delivered lectures and tutorial sessions across the full electrodynamics curriculum — electrostatics, special techniques (Laplace's equation, method of images, separation of variables, multipole expansion), electric fields in matter, magnetostatics, magnetic fields in matter, and electrodynamics.
 - **Curriculum Innovation & Student Outcome:** Identified that the set texts assumed mathematical prerequisites many students lacked; developed supplementary materials drawing on multiple textbooks to cover vector algebra, curvilinear coordinates, tensor and index notation, the Levi-Civita symbol and the Dirac delta function, which produced an increase in student assignment scores.
 - **Assessment & Feedback:** Prepared and graded exercise solutions and chapter quizzes; evaluated student assignments and examination scripts with detailed written feedback.
 - **Reference Texts:** Course based on D.J. Griffiths, *Introduction to Electrodynamics*, and J.D. Jackson, *Classical Electrodynamics*.
- **National Tsing Hua University** Hsinchu, Taiwan
Teaching Assistant — Mathematical Physics (P.h.D., Postgraduate & Undergraduate Physics) 2022 - 2023
 - **Course & Delivery Format:** Taught a one-semester Mathematical Physics course, delivering two classes per week (two-hour sessions) combining lectures and tutorial sections for students from diverse international backgrounds.
 - **Topics Delivered:** Covered the mathematical-methods syllabus, including probability theory and random variables (distributions, moments and generating functions, Central Limit Theorem), linear vector spaces and operator methods, Fourier analysis, ordinary and partial differential equations with special functions, and complex analysis (full topic list in the Courses Taught section below).

- **Course Material Preparation:** Prepared lecture notes, problem sets, and worked examples guiding students in applying quantitative methods to physics problems.
- **Assessment & Examination:** Prepared assignment questions, marked and moderated student assignments, and evaluated examination papers, providing structured and constructive feedback consistent with assessment policy.
- **Individual Student Support:** Provided one-on-one consultations and individual sessions to reinforce student comprehension of mathematical and computational methods.

• National Tsing Hua University

Hsinchu, Taiwan

Research Supervisor — Master's and Exchange Students

2023 - 2024

- **Master's Student Supervision:** Supervised Chen Wang and Wanyon Hsiao through problem formulation, collider-physics data analysis, statistical inference, and scientific writing — constraining new-physics couplings at the LHC and HL-LHC — resulting in publication in Journal of High Energy Physics (Q1 journal, 2024).
- **Exchange Student Supervision:** Mentored Thong T.Q. Nguyen (Academia Sinica, Taiwan) during an exchange program on a Forward Physics Facility New Physics search, supporting independent research-skill and statistical-analysis development, leading to publication in Physical Review D (Q1 journal, 2023).
- **Statistical Analysis Training:** Trained supervisees in the full analysis pipeline for collider data — maximum-likelihood estimation, hypothesis testing, exclusion-limit setting, sensitivity analysis, and uncertainty quantification — implemented reproducibly in Python and Mathematica.
- **Research Guidance:** Provided structured guidance on research methodology, literature review, critical analysis, statistical problem-solving strategies, and academic integrity, preparing students for independent scholarly work.
- **Collaborative Learning:** Fostered a collaborative research environment encouraging peer-to-peer learning, teamwork, and knowledge sharing among students with diverse academic backgrounds.

• National Tsing Hua University

Hsinchu, Taiwan

Organiser — Departmental Seminar & Postgraduate Workshop Series

Fall 2023

- **Seminar Series:** Organised the Physics Department seminar series, inviting external expert speakers, coordinating schedules, and arranging facilities.
- **Postgraduate Workshop Series:** Organised a Master's and Ph.D. workshop series for departmental postgraduate students, and submitted research funding proposals to support academic activities and international travel.

COURSES TAUGHT & TOPICS DELIVERED

- **Electrodynamics — Lectures & Tutorial Sessions:** Mathematical prerequisites (vector algebra, differential and integral calculus, curvilinear coordinates, tensor and index notation, Levi-Civita symbol, Dirac delta function, theory of vector fields); electrostatics; special techniques (Laplace's equation, method of images, separation of variables, multipole expansion); electric fields in matter (polarization, dielectrics); magnetostatics (Lorentz force law, Biot-Savart law, magnetic vector potential); magnetic fields in matter; electrodynamics (electromotive force). *Texts:* D.J. Griffiths, *Introduction to Electrodynamics*; J.D. Jackson, *Classical Electrodynamics*.
- **Mathematical Physics — Lectures & Tutorials:** *One-semester course; two 2-hour classes per week.* Vectors and tensors (vector calculus, index notation); linear vector spaces and operators (Dirac notation, inner products, Gram-Schmidt orthonormalisation, matrix algebra, similarity transforms, diagonalization, Hermitian and unitary operators, generalized functions, Dirac delta function); Fourier series, integrals and transforms; ordinary differential equations (power-series solutions, Sturm-Liouville problems, Hermite/Legendre/Laguerre/Bessel functions, spherical harmonics, Gamma/Beta/error functions); partial differential equations (Laplace, Poisson, Helmholtz; separation of variables); **probability theory and random variables** (probability distributions and densities, discrete and continuous distributions, moments and generating functions, Central Limit Theorem); complex analysis (analytic functions, Cauchy-Riemann conditions, singularities, residue theorem, contour integration). *Texts:* Arfken & Weber; Riley, Hobson & Bence; Boas.
- **Applied Statistical Methods — Postgraduate Research Supervision:** Delivered through one-on-one supervision rather than formal classroom teaching: maximum-likelihood and Bayesian inference, hypothesis testing, exclusion-limit setting, sensitivity analysis, uncertainty quantification, and reproducible statistical computing in Python and Mathematica — training three postgraduate students through to Q1 publication.
- **Laboratory & Experimental Training (as a student):** Completed General Physics Laboratory and Electronics Laboratory (B.Sc.) and Physics Laboratory (M.Sc.) coursework; has not yet held responsibility for coordinating or demonstrating undergraduate teaching laboratories.

TEACHING INTERESTS

- **Core Physics:** Electromagnetism, Classical Mechanics, Mathematical Physics, Quantum Mechanics, Statistical Physics.
- **Advanced / Upper-Level:** Particle Physics, Relativity, Cosmology, Astrophysics, Computational Physics.
- **Research & Computational:** Scientific Computing, Numerical Methods, Data Analysis and Statistical Methods for Physics.

TEACHING SKILLS & COMPETENCIES

- **Physics Content Expertise:** Electrodynamics (Griffiths, Jackson), classical mechanics and field theory foundations, mathematical physics, quantum and statistical methods underpinning collider phenomenology, general relativity and cosmology (B.Sc. thesis on general relativity, black holes and Kruskal diagrams; M.Sc. thesis on Higgs inflation; doctoral and postdoctoral work on inflation, primordial black holes and gravitational waves)
- **Mathematical Content Expertise:** Calculus, linear algebra, differential equations, probability and statistics, complex analysis, numerical methods, and mathematical modeling, taught at undergraduate and postgraduate level
- **Assessment Design:** Developing assessments, marking and moderating student work, providing constructive written feedback, applying assessment policy consistently, and evaluating learning outcomes
- **Pedagogical Skills:** Active learning strategies, inquiry-based and problem-based learning, scaffolded instruction, formative assessment, learner-centered and inclusive pedagogy
- **Diverse Student Support:** Taught students from varied cultural backgrounds (Taiwan, Korea, India, and international cohorts); experience adapting explanation and pacing to individual learning needs
- **Technology Integration:** eLearning platforms, learning management systems, Python and Mathematica for live mathematical demonstration, and self-built interactive visualisation tools (Inflation Explorer, VizDash Pro) used to make abstract models visible to students
- **Curriculum Development:** Contributed to learning needs analysis, course content development, learning outcomes, resource preparation, and assessment method refinement in collaboration with teaching teams
- **Communication Skills:** Explaining complex physical and mathematical concepts clearly to diverse audiences; 20+ conference and seminar presentations to specialist and non-specialist audiences
- **Collaboration & Teamwork:** Working collaboratively with teaching teams, seeking peer feedback, participating in course reviews, and contributing to school activities and planning
- **Professional Commitment:** Continuous professional development, maintaining currency in teaching practice and subject expertise, and engaging with professional communities

FUTURE RESEARCH PROGRAMME

- **Primordial black holes and early-universe cosmology:** Develop theoretical predictions for PBH formation, Hawking radiation, gamma-ray signatures and associated gravitational-wave signals.
- **New physics and astrophysical probes:** Investigate dark photons, axion-like particles and sterile neutrinos through collider and astrophysical observations.
- **Particle physics–astrophysics interface:** Develop interdisciplinary collaborations connecting theoretical particle physics with astrophysics, cosmology and space science.

These projects are computationally driven and can support Honours, Master's and PhD research projects.

EDUCATION

- **National Tsing Hua University** Hsinchu, Taiwan
Ph.D. in Physics; Thesis: New Physics Search at Forward Physics Facility of LHC 09/2019 - 07/2024
- **Indian Institute of Technology Indore** Indore, India
M.Sc. in Physics; Thesis: Higgs Inflation 07/2016 - 07/2018
- **University of Kerala** Thiruvananthapuram, India
B.Sc. in Physics; Thesis: General Relativity, Black Holes, and Kruskal Diagram 07/2012 - 07/2015

RESEARCH EXPERIENCE

- **Senior Research Scientist** Seoul, Korea
Seoul National University of Science and Technology 09/2024 - 08/2025
 - **Independent Research Agenda:** Developed and validated mathematical models for complex physical systems using Python, numerical methods, and statistical frameworks, sustaining an independent line of enquiry in theoretical physics and cosmology
 - **Publication Output:** Produced peer-reviewed output in internationally refereed journals, including work on the neutrino dipole portal and primordial black hole gamma-ray signatures
 - **Statistical Analysis & Inference:** Applied Bayesian inference, time series analysis, hypothesis testing, likelihood analysis, and confidence interval estimation to analyse complex datasets and quantify uncertainties
 - **Python Programming & Data Analytics:** Designed computational pipelines for large-scale data processing (>10TB), statistical analysis, and scientific visualization using Python (NumPy, SciPy, Pandas, Matplotlib, Seaborn) in Linux and Mac environments
 - **Research Collaboration & Leadership:** Led multiple international research collaborations across 5+ countries, coordinating projects with diverse colleagues, managing timelines, and delivering research milestones
 - **Research Funding & Proposal Development:** Secured competitive research grants through strategic proposal development, articulating scientific value and engaging funding bodies

- **Mentorship & Knowledge Transfer:** Supervised and mentored early-career researchers and graduate students on computational techniques, data analysis methodologies, and research best practices
- **Model Validation & Uncertainty Quantification:** Translated theoretical frameworks into validated numerical simulations, testing models against observational constraints and managing uncertainty in predictions

- **Senior Visiting Research Scientist** Adelaide, Australia
University of Adelaide 03/2025 - 05/2025
 - **Interdisciplinary Research Collaboration:** Collaborated with multidisciplinary academic teams on research projects involving statistical modeling, data analysis, and quantitative problem-solving
 - **Computational Statistics & Data Modeling:** Conducted research involving computational statistics, data simulation, and numerical modeling using Python and Mathematica for complex system analysis
- **Doctoral Researcher** Hsinchu, Taiwan
National Tsing Hua University 09/2019 - 07/2024
 - **Scientific Writing & Publication:** Published 15+ peer-reviewed articles in high-impact Q1 journals, demonstrating scientific documentation, technical reporting, and communication of complex analytical methods
 - **Computational Modeling & Simulation:** Developed mathematical models and numerical algorithms for analyzing complex physical systems, with strong emphasis on data-driven modeling and predictive analytics
 - **Advanced Statistical Methods:** Applied hypothesis testing, confidence interval estimation, likelihood analysis, regression techniques, and model selection methods to research problems with quantifiable uncertainty
 - **Algorithm Development & Numerical Methods:** Designed and optimized numerical algorithms for solving differential equations, conducting Monte Carlo simulations, and implementing optimization procedures
 - **High-Performance Computing:** Utilized HPC systems and Linux environments to execute computationally intensive simulations and parallel processing workflows for large-scale data analysis
 - **Large-Scale Data Processing & Management:** Designed and implemented data processing workflows for handling, analyzing, and visualizing multi-terabyte datasets using Python and specialized analytical software
 - **Literature Review & Knowledge Synthesis:** Conducted comprehensive literature reviews to identify research gaps, synthesize findings from diverse sources, and inform evidence-based research strategies

ACADEMIC VISITS & INTERNSHIPS

- **University of Adelaide** Adelaide, Australia
Host: Prof. Anthony Gordon Williams Mar 2025 - May 2025
- **University of Notre Dame** Notre Dame, Indiana, USA
Host: Prof. Yuhsin Tsai Mar 2024 - Apr 2024
- **National Tsing Hua University** Hsinchu, Taiwan
Host: Prof. Tomotsugu Goto (internship) Mar 2019 - Aug 2019
- **Indian Institute of Technology Guwahati** Guwahati, Assam, India
Host: Prof. Debaprasad Maity Feb 2019

PROJECTS

- **Quantitative System Simulations**
Executed 4 HPC projects involving cosmological simulations and analytical modeling of early-universe phenomena.
 - **Objective:** Developed theoretical and numerical models to study early-universe phenomena including inflation, primordial black holes (PBHs), and gravitational waves (GWs).
 - **Model Building & Simulation:** Formulated cosmological models through analytical derivations; implemented numerical simulations using Python and Mathematica when closed-form solutions were intractable.
 - **Algorithm Development:** Designed and executed complex algorithms to compute gravitational wave signals and PBH mass spectra based on early-universe dynamics.
 - **Data Analysis & Visualization:** Processed simulation outputs to analyze parameter dependencies and visualize physical observables. Created plots and heatmaps to interpret trends and constraints.
 - **Tools & Technologies:** Python, NumPy, SciPy, Matplotlib, Mathematica; proficient in symbolic and numerical computation for high-precision modeling.
 - **Outcome:** Contributed to peer-reviewed publications; models provided new insights into GW production and PBH formation mechanisms.
- **Data-Driven Modeling Projects**
Completed 12 high-performance computing (HPC) projects focused on simulation, modeling, and data analysis in collider physics.
 - **Objective:** Designed data-driven strategies to identify anomalies, new physics at experiments like LHC, DUNE, and FPF using simulated and theoretical data to constrain model parameters.
 - **Data Simulation & Modeling:** Built analytical and numerical frameworks to convert models into Monte Carlo simulations using MadGraph, Pythia, and ROOT. Simulated rare processes and managed large-scale event generation.

- **Statistical Analysis:** Applied likelihood-based methods, profile ratios, and confidence intervals to estimate exclusion regions, conduct sensitivity projections, and perform hypothesis testing.
- **Data Engineering:** Processed high-dimensional datasets, performed feature engineering on kinematic variables, and developed data pipelines for extraction, transformation, and selection.
- **Visualization & Reporting:** Created histograms, ROC curves, correlation matrices, heatmaps, contour plots, lego plots, Dalitz plots, stacked histograms, invariant mass distributions, limit plots with Brazil bands, scatter plots, surface plots, error bar plots, and cut-flow tables with Python, Matplotlib, and ROOT; communicated results through clear, data-driven reports.
- **Tools & Technologies:** Python, Pandas, NumPy, SciPy, Matplotlib, Seaborn, ROOT, Scikit-learn, TensorFlow, MadGraph, Pythia, Mathematica, LaTeX; familiar with HPC systems for large-scale processing.
- **Outcome:** Published in Q1 journals; methods adopted in experimental analyses for data filtering and strategy design.

TEACHING & RESEARCH SOFTWARE PROJECTS

• Inflation Explorer — Interactive Cosmological Visualisation Tool

github.com/ouseph444/Inflation-Explorer | *Python, Plotly Dash, NumPy, SciPy*

- **Objective:** Built an interactive browser-based dashboard for exploring inflationary cosmology models, enabling real-time parameter sweeps of slow-roll potentials with immediate visual feedback on observational predictions; serves as both a research communication tool and a graduate-level teaching aid.
- **Interactive Dashboard Design:** Constructed a multi-panel Plotly Dash dashboard with slider-driven parameter controls; users tune inflationary potential parameters (mass, coupling constants) and instantly observe effects on the scalar power spectrum, spectral index n_s , and tensor-to-scalar ratio r ; all panels update dynamically without page reload.
- **Numerical Modelling & Computation:** Implemented numerical integration of the Klein-Gordon and Friedmann equations using `scipy.integrate.odeint`; computed slow-roll parameters ϵ and η analytically and cross-validated against full numerical solutions; derived CMB observables (n_s , r) and overlaid Planck 2018 and BICEP/Keck experimental constraint bands as shaded exclusion regions.
- **Physics Communication:** Designed the tool to bridge theoretical derivations and visual intuition, making inflationary model comparison accessible to non-specialist researchers and graduate students; includes annotated tooltips explaining the physical meaning of each adjustable parameter and output observable.
- **Technical Stack:** Python 3, NumPy, SciPy (`odeint`, `quad`), Plotly Dash, Plotly Express, Matplotlib (static export), Jupyter Notebook (prototype and documentation).

• VizDash Pro — Scientific Visualisation & Statistical Analysis Studio

ouseph444.github.io/VizDash-Pro- | *JavaScript, Plotly.js, PapaParse, MathJax, KaTeX, Tailwind CSS*

- **Overview:** Designed and developed a browser based scientific visualisation and statistical analysis platform for researchers and data scientists, providing publication quality plotting, interactive dashboards, and fully client side, privacy preserving data processing.
- **Data Analysis:** Implemented statistical tools including linear and polynomial regression, hypothesis testing (t test and ANOVA), Bayesian inference, confidence interval estimation, and likelihood visualisation for exploratory and quantitative analysis.
- **Visualisation:** Built a modular plotting engine supporting more than 30 chart types, including statistical, multivariate, and 3D visualisations, with regression overlays, confidence intervals, annotations, and LaTeX mathematical rendering.
- **Engineering:** Developed a robust CSV, TSV, and JSON data ingestion pipeline with automatic data profiling, validation, interactive dashboards, drag and drop layouts, and export to publication ready PNG, SVG, and HTML reports.
- **Technology:** JavaScript, Plotly.js, PapaParse, Sortable.js, Tailwind CSS, MathJax, KaTeX, HTML, CSS; fully offline capable with no backend dependencies.

• Personal Academic Webpage

ouseph444.github.io | *HTML, CSS, JavaScript, GitHub Pages*

- **Research Showcase & Discoverability:** Integrated live hyperlinks to InspireHEP author profile, Google Scholar, LinkedIn, and GitHub; embedded downloadable CV link and direct email contact; structured all 15+ peer-reviewed publication entries with DOI and arXiv links for one-click access.
- **Design & Deployment:** Designed and deployed a fully responsive single-page academic portfolio hosted on GitHub Pages; structured into biography, research overview, publication list, and contact sections, following modern web-design conventions with a clean professional aesthetic.
- **Technical Stack:** HTML5, CSS3 (custom responsive grid, media queries), Vanilla JavaScript, GitHub Pages, Font Awesome icons.

TECHNICAL SKILLS

- **Programming Languages:** Python (advanced - 5+ years), R, SQL, Wolfram Mathematica (advanced), JavaScript
- **Physics & Research Software:** Monte Carlo simulation frameworks (MadGraph, Pythia), data analysis software (ROOT, Delphes), symbolic computation (FeynRules, FeynCalc, FeynArts)
- **Python Expertise:** NumPy, SciPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow; scientific computing, data analysis, and spatial data processing
- **Mathematical & Numerical Modeling:** Numerical simulations, differential equations, Monte Carlo methods, optimization algorithms, predictive modeling, sensitivity analysis, parameter estimation
- **Statistical Analysis Methods:** Bayesian inference, time series analysis, hypothesis testing, regression analysis (linear, non-linear, multivariate), likelihood analysis, confidence intervals, model selection and validation
- **Data Visualization & Teaching Tools:** Matplotlib, Seaborn, Plotly Dash, Plotly.js, Power BI, Mathematica, Excel; interactive dashboards and publication-quality figures for technical and student audiences
- **High-Performance Computing:** Linux environments (advanced - 5+ years), parallel computing, computational optimization, batch processing systems, shell scripting
- **Technical Documentation:** LaTeX (expert), Markdown, scientific writing, technical reporting, documentation standards, proposal development for research funding
- **Operating Systems:** Linux (proficient - 5+ years daily use), Mac, Windows
- **Languages:** English (Fluent), Malayalam (Native), Tamil (Intermediate), Hindi (Intermediate)

GRANTS, FUNDING & RECOGNITION

- **Major Scholarships:** 5-year NTHU International Ph.D. Scholarship (2019-2024, highly competitive); Taiwan Experience Education Program (TEEP) Scholarship (2019)
- **Research Grants:** Secured multiple competitive travel grants for international conferences and collaborative research demonstrating ability to articulate scientific value and engage funding bodies
- **Grant Writing Experience:** Demonstrated expertise in proposal development, project planning, budget justification, and stakeholder engagement for successful funding acquisition
- **Research Impact:** 15+ peer-reviewed publications in Q1 high-impact journals; strong citation record and international recognition in research community
- **Academic Excellence:** Ranked top 6% (694/10,989) in All-India Joint Admission Test for Masters (JAM) Physics Examination (2016)
- **Professional Certifications:** PTE Academic certification (2023), TOEIC certification (2022) demonstrating professional English proficiency

PUBLICATIONS

Peer-Reviewed Journal Articles

1. S. K. Kang and **C.J. Ouseph**, "Neutrino Dipole Portal," Int. J. Mod. Phys. A, vol. 41, no. 02, p. 2530022, 2026. [arXiv:2510.18181 [hep-ph]].
2. **C.J. Ouseph**, Giorgio Busoni, John Gargalionis, Sin Kyu Kang, Anthony G. Williams "Probing Double-Peaked Gamma-Ray Spectra from Primordial Black Holes with Next-Generation Gamma-Ray Experiments," arXiv:2507.16244, 2025
3. K. Cheung, **C.J. Ouseph**, P.-Y. Tseng, and S. K. Kang, "Searching for Dark Photon Tridents Through Primordial Black Hole Signatures," Mar. 2025. arXiv: 2503.04175 [hep-ph].
4. K. Cheung, Y. Kim, Y. Kwon, **C.J. Ouseph**, A. Soffer, and Z. S. Wang, "Probing dark photons from a light scalar at Belle II," JHEP, vol. 05, p. 094, 2024. DOI: 10.1007/JHEP05(2024)094. arXiv: 2401.03168 [hep-ph].
5. Athron, Y. Bai, S. S. Bao, D. Barducci, X. J. Bi, T. Cai, L. Calibbi and J. Cang, **C.J. Ouseph et al.** "New Physics Search at the CEPC: a General Perspective," doi:10.1088/1674-1137/ae1194 [arXiv:2505.24810 [hep-ex]].
6. K. Cheung, C.-T. Lu, **C.J. Ouseph**, and P. Sarmah, "Exploring interference effects between two ALP effective operators at the LHC," JHEP, vol. 09, p. 101, 2024. DOI: 10.1007/JHEP09(2024)101. arXiv: 2404.14833 [hep-ph].
7. K. Cheung and **C.J. Ouseph**, "Interpretation of excess in $H \rightarrow Z\gamma$ using a light axionlike particle," Phys. Rev. D, vol. 110, no. 5, p. 055016, 2024. DOI: 10.1103/PhysRevD.110.055016. arXiv: 2402.05678 [hep-ph].
8. K. Cheung, **C.J. Ouseph**, and S. K. Kang, "Unveiling the Invisible: ALPs and Sterile Neutrinos at the LHC and HL-LHC," JHEP, vol. 04, p. 024, 2025. DOI: 10.1007/JHEP04(2025)024. arXiv: 2412.08212 [hep-ph].
9. K. Cheung, **C.J. Ouseph**, and P.-Y. Tseng, "NANOGrav and other PTA signals and PBH from the modified Higgs inflation," Eur. Phys. J. C, vol. 84, no. 9, p. 906, 2024. DOI: 10.1140/epjc/s10052-024-13268-6. arXiv: 2307.08046 [hep-ph].

10. K. Cheung and **C.J. Ouseph**, "Axionlike particle search at Higgs factories," Phys. Rev. D, vol. 108, no. 3, p. 035003, 2023. DOI: 10.1103/PhysRevD.108.035003. arXiv: 2303.16514 [hep-ph].
11. K. Cheung and **C.J. Ouseph**, "Constraining the transitional magnetic dipole moment interaction between active and heavy neutrinos with the Z' at FASER ν ," Eur. Phys. J. C, vol. 83, no. 7, p. 593, 2023. DOI: 10.1140/epjc/s10052-023-11784-5. arXiv: 2205.11077 [hep-ph].
12. K. Cheung and **C.J. Ouseph**, "Sensitivities on dark photon from the forward physics experiments," JHEP, vol. 10, p. 196, 2022. DOI: 10.1007/JHEP10(2022)196. arXiv: 2208.04523 [hep-ph].
13. K. Cheung, **C.J. Ouseph** and T. Wang, "Non-standard neutrino and Z' interactions at the FASER ν and the LHC," JHEP, vol. 12, p. 209, 2021. DOI: 10.1007/JHEP12(2021)209. arXiv: 2111.08375 [hep-ph].
14. **C.J. Ouseph** and K. Cheung, "Higgs Inflation with four-form couplings," J. Phys. G, vol. 48, no. 5, p. 055001, 2021. DOI: 10.1088/1361-6471/abefa4. arXiv: 2002.12010 [hep-ph].

Complete Publication List: <https://inspirehep.net/authors/1842258>

SELECTED PUBLICATIONS (DEMONSTRATING SUPERVISION SUCCESS)

1. K. Cheung, W. Hsiao, **C.J. Ouseph**, and C. Wang, "Probing the gauge-boson couplings of axion-like particle at the LHC and high-luminosity LHC," JHEP, vol. 05, p. 324, 2024. DOI: 10.1007/JHEP05(2024)324. *(Co-authored with supervised Master's students Chen Wang and Wanyon Hsiao)*
2. K. Cheung, T. T. Q. Nguyen, and **C.J. Ouseph**, "Leptoquark search at the Forward Physics Facility," Phys. Rev. D, vol. 108, no. 3, p. 036014, 2023. DOI: 10.1103/PhysRevD.108.036014. *(Co-authored with supervised exchange student Thong T.Q. Nguyen)*

CONFERENCE PRESENTATIONS

- **April 2025:** "Searching for Dark Photon Tridents Through Primordial Black Hole Signatures," Seminar talk, National Tsing Hua University, Taiwan.
- **March 2025:** "Searching for Dark Photon Tridents Through Primordial Black Hole Signatures," Seminar talk, University of Adelaide, Australia.
- **Oct 2024:** "New Physics Search at FPF LHC, HL-LHC, and Higgs Factories," Invited talk, Konkuk University, South Korea.
- **June 2024:** "New Physics Search at the Forward Physics Facility of LHC," Seminar talk, Academia Sinica, Taiwan.
- **April 2024:** "Probing dark photons from a light scalar at Belle II," Seminar talk at National Yang Ming Chiao Tung University, Taiwan.
- **March 2024:** "New Physics Search at the Forward Physics Facility of LHC," Seminar talk, University of Notre Dame, Indiana, United States.
- **June 2023:** "Sensitivities on dark photon from the Forward Physics Experiments," Workshop: The Future is Non-perturbative, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.
- **April 2023:** "Implications of NSI constraints from ANTARES and IceCube on a simplified Z' model," Journal Club talk, National Tsing Hua University, Taiwan.
- **March 2023:** "New Physics Search at the Forward Physics Facility of LHC," Seminar talk, National Taiwan Normal University, Taiwan.
- **October 2022:** "BSM Physics at FASER ν from neutrino nucleon scattering," NuINT 2022 - The 13th International Workshop on Neutrino-Nucleus Interactions in the Few GeV Regions, National Seoul University, Korea.
- **June 2022:** "BSM Physics at FASER ν ," Workshop: The Future is Illuminating, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.
- **January 2022:** "Nonstandard Neutrino and Z' interaction at FASER ν ," Discussion section, Neutrino-Nucleus Interactions in the Standard Model and Beyond, CERN, Geneva, Switzerland.
- **June 2021:** "Neutrino Scattering at FASER ν and Search for Z' ," Workshop: The Future is DARK, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.

PROFESSIONAL ATTRIBUTES & ADDITIONAL INFORMATION

- **Residency & Relocation:** Australian Permanent Resident currently based in Adelaide, South Australia; eligible to live and work in New Zealand under the Trans-Tasman Travel Arrangement and available to relocate to Wellington
- **Communication Skills:** Communicating complex scientific concepts to technical and non-technical audiences through publications, 20+ conference and seminar presentations, technical reports, and stakeholder engagement
- **Collaboration Experience:** International research collaborations across diverse teams; experience in multidisciplinary environments and engaging with colleagues from academia, government, and industry
- **Cultural Competency:** Worked and taught in Taiwan, Korea, India and Australia; committed to inclusive teaching and to learning and upholding the University's obligations under Te Tiriti o Waitangi
- **Adaptability & Learning:** Rapid acquisition of new technical skills and application of existing expertise to novel domains; strong computational foundation enables quick proficiency with new tools and course material
- **Background Checks:** Willing to obtain any required background checks and clearances
- **Languages:** English (Fluent - professional working proficiency), Malayalam (Native), Tamil (Intermediate), Hindi (Intermediate)

REFERENCES

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Prof. Tseng, Po-Yen

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